

Synopsis: Solar-System Null Test of the Hysterical Response

What the bound says (without the orbital algebra)

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This is a short guide to the Solar-System null test of the Hysterical Response. It assumes the response-theory and galactic-modelling papers. Each section says *why* the claim matters, then states the result. Orbital algebra and observational detail live in the full paper.

1 Why a Solar-System null test

Galactic modelling supplies $g_{\text{resp}} = CQ/r$ for response-supported systems. That does not force the Sun into the same regime: the Solar amplitude $V_{\odot}^2 = CQ_{\odot}$ is a separate, system-dependent quantity. The question here is only how large $V_{\odot}$ can be before Saturn-scale ephemerides object.

2 Point-source specialization

Outside a localized Solar source, the same far-field law reads

$$g_{\text{resp}} = \frac{V_{\odot}^2}{r}, \quad \epsilon(r) = \frac{V_{\odot}^2 r}{GM_{\odot}}.$$

The fractional correction grows with radius, so outer planets bite hardest. An equivalent enclosed mass is $M_{\text{resp}}(< r) = V_{\odot}^2 r/G$.

3 Perihelion signature

A purely radial $1/r$ acceleration produces a retrograde perihelion drift.

Proposition 3.1 (Perihelion shift of a logarithmic response).

$$\Delta\varpi = -\frac{2\pi V_{\odot}^2 a}{GM_{\odot}} \frac{\sqrt{1-e^2}}{1+\sqrt{1-e^2}}$$

per orbit, reducing to $-\pi V_{\odot}^2 a/GM_{\odot}$ for $e \rightarrow 0$.

(The Gauss reduction to the true-anomaly integrand is a literature input; the remaining integral algebra is Lean-checked.)

4 Saturn ceiling

Translating the Pitjeva–Pitjev Saturn perihelion uncertainty $4.7 \times 10^{-6} \text{ arcsec yr}^{-1}$ through that formula yields

$$V_{\odot} < 0.141 \text{ m s}^{-1}$$

at the quoted one-sigma level. An enclosed-mass consistency scale from the same literature sits near $V_{\odot} \lesssim 0.086 \text{ m s}^{-1}$.

5 Benchmark inside the window

For $V_{\odot} = 0.01 \text{ m s}^{-1}$, Saturn sees $g_{\text{resp}} \sim 7 \times 10^{-17} \text{ m s}^{-2}$ and $\dot{\varpi}_{\text{resp}} \sim -2.4 \times 10^{-6} \text{ arcsec per century}$ —below the quoted limit. Cassini γ is an independent relativistic consistency condition, not a direct $V_{\odot}$ measurement. A Saturn-centric amplitude V_{Sat} needs a dedicated Grand Finale re-fit.

6 What is and is not claimed

The far-field shape is inherited. The perihelion formula is a theorem of the stated Gauss integrand. The numerical ceiling is an observational translation, not a detection. A nonzero subcritical window remains open; closing it requires explicit $V_{\odot}^2/r$ ephemeris fits.